

Technical Requirements Document

03 — Technical Requirements Document (TRD)

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1. Stack

Concern	Choice
Language	JavaScript (JSX), ES modules
UI	React 18
Build	Vite 5
Routing	react-router-dom v6
Styling	Tailwind CSS 3 (+ PostCSS, Autoprefixer)
Animation	framer-motion
Charts	recharts
Icons	react-icons
HTTP (future API)	axios
Exports	jspdf, jspdf-autotable, xlsx

All five tools shipped on this **same** stack, which is what made the merge clean.

2. Project structure (frontend)

```
frontend/
├─ index.html           # loads Inter, Plus Jakarta Sans, Outfit fonts
├─ vite.config.js       # @ → ./src alias, dev port 3000
├─ tailwind.config.js   # ONE merged theme (see §4)
├─ package.json         # ONE merged dependency set
├─ src/
│  ├─ main.jsx          # providers + BrowserRouter (the single router)
│  ├─ App.jsx → routes/AppRoutes.jsx
│  ├─ index.css          # global dark theme + Tailwind directives
│  ├─ pages/             # shell pages incl. <Tool>Page.jsx mount wrappers
│  ├─ components/ui/HubLink.jsx # back-to-hub pill used by every tool
│  ├─ context/, hooks/, layouts/, services/, lib/ # shell internals
│  └─ features/
│     ├─ power-planner/
│     ├─ reason-eliminator/
│     ├─ meeting/
│     └─ time-finder/
```

3. Routing conventions

- The shell owns the **only** `<BrowserRouter>` (in `src/main.jsx`).
- A tool with internal routes is mounted with a **splat**: `<Route path="/time-finder/*" element={<TimeFinderPage/>}/>`.
- Inside a tool, `<Route>` paths are **relative** (no leading `/`); `path=""` is the tool's home.
- Inside a tool, `navigate()` / `<Link to>` use **absolute app paths** that include the tool's mount prefix (e.g. `/meeting-framework/meeting-list`). This is why bare-path tools (Meeting, Time Finder) had their nav prefixed during the merge.
- Returning to the hub: `<HubLink/>` → `/home`.

4. Styling conventions

- **One** Tailwind config and **one** global CSS. The app shell is dark; tools render light inside their own full-screen wrapper.
- Token collisions were resolved by **namespacing**, not by overriding:
 - `brand` = shell numeric scale + tools' named keys (merged).
 - Meeting `ink` / `ink-soft` → `mkink` / `mkink-soft`.
 - Time Finder `ink-900/500/200` → `tfink-900/500/200`.
 - Time Finder `bg-neutral-50` → `bg-[#fafafa]` (shell repurposes `neutral` as a flat category color, which removes the default `neutral-*` scale).
- A tool needing component-level CSS classes ships `features/<tool>/scoped.css` (plain rules + `@apply`, **no** `@layer` wrapper — each file compiles alone).
- **Rule of thumb for adding a tool**: never globally import a tool's `index.css` (it carries a `body` reset). Wrap the tool and namespace any custom color tokens that clash with the shell or another tool.

5. Aliases

- `@` → `frontend/src` (Vite + jsconfig).
- Reasons Eliminator originally used `@/` for its own root; during the merge those were rewritten to `@/features/reason-eliminator/...`. Other tools use relative imports.

6. Build, run, deploy

```
cd frontend
npm install
npm run dev      # http://localhost:3000
npm run build    # → frontend/dist
npm run preview  # serve the production build
```

- Deploy `frontend/dist` to any static host with SPA fallback to `index.html`.
- Build currently emits a chunk-size warning (jspdf/recharts are large). Optional follow-up: `manualChunks` to split vendor bundles. Not blocking.

7. Known technical risks / follow-ups

Risk	Note	Severity
Time Finder generic <code>localStorage</code> keys	<code>assessments</code> , <code>currentAssessment</code> , etc. are not namespaced. No clash today, but namespace them (<code>timefinder.*</code>) before adding any tool that could reuse those names.	low
Meeting font	Tool components use <code>font-sans</code> (now Inter); the wrapper sets <code>font-meeting</code> (Outfit). Most text renders Inter. Cosmetic only.	low
Light/dark theme toggle	The shell can switch to a light theme; tools assume the default dark <code>html</code> for namespaced <code>ink</code> resolution. Tools set their own surfaces, so impact is minor, but verify if the theme toggle is exposed to users.	low
Bundle size	<code>jspdf</code> + <code>recharts</code> inflate the largest chunks.	low

8. Quality gates

- ☒ `npm run build` passes (369 modules, all tool chunks emitted).
- ☐ Browser QA per tool (navigate hub → each tool → back; create/edit/persist).
- ☐ Lint pass (no shared eslint config yet across features).